

A Prime–Archimedean Formulation Atlas

Valuation simplices, constant-producing paths, situated surfaces, and the completed zeta seam

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23 July 2026 – first exact atlas cut

Abstract. This paper constructs the prime–archimedean formulation atlas requested by the laboratory. It is not a gallery of formulas grouped only because they return the same scalar. Its horizontal geometry is the exact exponent lattice of finite prime places: degree fibers are integer simplices, degree-preserving moves form the root lattice A_{r-1} , and prime powers are repeated traversal of primitive axes. Its vertical geometry consists of declared formulation changes: Gaussian factorization and arctangent addition for π ; factorial valuations, limits, continued fractions, differential flow, and the exponential cover for e ; and Fourier–Poisson–Mellin completion for zeta. The finite and archimedean directions already meet exactly in the product formula for rational numbers and, analytically, in the completed zeta function. A bounded exact construction now resolves every term of three arctangent arms and one Chudnovsky recurrence into named factor channels, signed divisors, internal annihilation, additive cancellation, and the surviving partial-sum face. It shows that a recurrence is a moving carrier rather than a retained factor archive, and that equal transcendental return does not conserve one finite-prime support. A pathwise Riemannian mean-transport lemma supplies the complementary continuous law: smooth deformation integrates within a situated path, while branch, rank, phase, and valuation seams contribute explicit discrete terms. A second bounded exact construction joins transformed-grid germs, completed-zeta symmetry, logarithmic rebase, and centered Smith incidence. It distinguishes a regular crossing, zero, critical branch, and distinct-source coincidence; derives $\frac{1}{2}$ as the fixed projector of $s \mapsto 1 - \bar{s}$; and carries the same signed normal coordinate as circle-versus-spiral conduct and unit-shell displacement, without floating evaluation. A third bounded exact construction turns the reciprocal Smith pair into one moving chord. It carries the chord’s complex-plane intersection, oriented axis, two pole paths, common and coupling metrics, and triangular holonomy separately. Exact rational precession moves both the intersection and the stereographic direction by one complex phase; a coupled edge produces an oriented pole imbalance which reverses under pole exchange. A conic-pencil discriminant remains a separate typed seam. A further purpose is to explain why Wolfram MathWorld’s ray-traced mathematical objects are germane. The Bowl of Integers, Cayley cubic, Chen–Gackstatter family, and Klein quartic each place a compact local law inside a globally nontrivial orbit, singularity, period problem, or quotient. Ray tracing is formalized as a situated receiver rather than an outside camera. The image testifies about one real affine cut while concealing other roots, complex and projective faces, interior sheets, and resolutions. The paper ends at one precise Riemann-Hypothesis edge: the local charts now agree exactly that a nontrivial zero mode is on the critical seam iff its normal exponent vanishes, its rebase is unitary, and its Smith image lies on the unit shell. The remaining global construction must exclude every nontrivial off-shell zero mode while covariantly retaining

completed response, adjoint, boundary, and holonomy. The contribution is an exact census, a concrete two-direction atlas, several nontrivial populated cells, and a bounded program for that transport law.

Contents

I	The question and the object constructed	4
I.1	Claim grades	5
II	The finite-prime geometry	5
II.1	Prime support is an exponent lattice	5
II.2	Exact group counts per prime axis	6
II.3	The measured finite aperture	7
II.4	Residue loops are a different fiber	8
III	Horizontal transport: roots, fractions, and recurrence	8
III.1	Degree-preserving prime moves	8
III.2	Continued fractions are rebase words	9
III.3	The flat triangle and the possible residual	9
IV	Vertical transport: formulations of π and e	9
IV.1	A formulation is a path, not merely its limit	9
IV.2	A complete π cell: Machin through Gaussian factorization	10
IV.3	A second π cell: a character field over odd primes	10
IV.4	The e family carries factorial prime currents	11
IV.5	A series has a term current and an accumulation current	12
IV.5.1	Exact bounded presentation-topology cut	13
IV.5.2	Chudnovsky is the same carrier law at higher arity	14
IV.5.3	One return, nonidentical finite-prime charts	14
IV.6	The π - e seam is exponential covering geometry	14
V	The exact prime-archimedean bridge	15
V.1	The product formula	15
V.2	Situated mean transport: the continuous companion to a discrete event	15
V.3	Zeta completion adds the infinite place	17
V.4	The conformal rebase atlas: one normal hand in four charts	18
V.5	The reciprocal chord acquires a discrete connection	21
VI	The formulation atlas as a bicategory with an exact rechart core	23
VI.1	Objects, arrows, and two-cells	23
VI.2	Compression in the atlas	25
VII	Ray tracing as a situated mathematical receiver	26
VII.1	Why the gallery is relevant	26
VII.2	The camera cannot leave the system	26
VIII	Four surface cells	27
VIII.1	Bowl of Integers: an integer orbit of tangent constraints	27
VIII.2	Cayley cubic: singular nodes as founding seams	28
VIII.3	Chen-Gackstatter: local complex data and global period closure	28
VIII.4	Klein quartic: the strongest prime-archimedean surface cell	28
IX	What the rest of the gallery contributes	29
X	Relation to the Riemann Hypothesis	30

X.1	What this atlas adds to the existing RH paper	30
X.2	Why infinitely many π - e formulas matter	30
X.3	The local equivalence and the global carrier	31
X.4	The next global square	32
XI	Bounded analytical program	32
XII	Conclusion	33
	Bibliography	35

I The question and the object constructed

The present question is not whether π , e , prime numbers, and attractive surfaces “look related.” It is whether their distinct formulations expose reusable transformation laws once the identity of each path, its receiver, and its unresolved residual are retained.

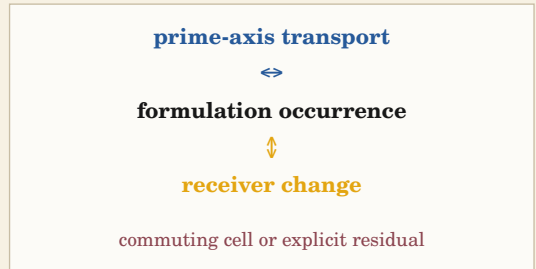
The standing evidence was insufficient in two ways. Earlier exact experiments counted prime-power apertures and bounded formula families, but they did not place those counts into one closed combinatorial geometry. Earlier Riemann work assembled Euler, theta, Mellin, and Weil faces, but it did not populate the vertical formulation direction with the concrete π – e cells or formalize a visual observation as a receiver event.

Geometric construction

The atlas constructed in this paper

Problem. Keep the exact prime composition of a formulation while allowing its analytic, algebraic, geometric, or rendered face to change. Determine which squares commute, which expose a residual, and what extra structure an RH-bearing square would have to preserve.

Construction and demonstration. The horizontal fibers are the exact valuation simplices $\Delta_{S,d}$. Their transitions are inclusions of prime support and degree-preserving A_{r-1} transfers. Vertical transitions include Gaussian norm and argument, logarithm and exponential, Fourier and Poisson, Mellin and Gamma completion, quotient maps, and situated renderings. A cell closes only when both paths return the same addressed occurrence with the same declared structure.



Algebraic face.

$$(S, \alpha, F) \xrightarrow{\text{prime}} (T, \beta, F') \quad \text{and} \quad (S, \alpha, F) \xrightarrow{\text{formulation}} (S, \alpha, G).$$

The square compares “prime then formulation” with “formulation then prime.” Its residual is the difference between those two composite transition laws, not the difference between two rounded output digits.

Therefore. Equal returned values do not erase nonidentical paths. Conversely, a visually different face need not be a different intrinsic object. The atlas retains both statements at once.

The stopping condition for this cut is a populated, proof-auditable atlas with an exact next RH-bearing question. No Soma engine run and no image mining are part of the construction.

I.1 Claim grades

Grade	Use in this paper
Classical identity	Factorization, stars and bars, the product formula, Gaussian norms, arctangent addition, Fourier/Poisson/Mellin transport, and the standard surface invariants are used under their ordinary hypotheses.
Exact laboratory synthesis	Several classical objects are placed in one two-direction atlas, and the exact prime-axis interior of selected π - e formulations is retained.
Bounded measurement	Earlier finite observations through prime 509, the 82 prime-power gears through 313, and the degree-two and degree-three formula apertures are recounted only inside their declared cutoffs. The exact presentation-topology cut additionally carries 12 terms of each arctangent arm and eight Chudnovsky terms as complete divisor-and-accumulation paths. The conformal-rebase cut carries four exact local germs, one retained distinct-preimage coincidence, four completed-zeta orbits, four log-polar rebase paths, and four centered Smith rows. The moving-chord cut additionally carries one exact Smith tilt and precession, three pole-edge foils, two induced sheet metrics, one flat and one curved connection face, and four conic-pencil phases. Those populations establish their local laws, not a global zero census.
Open construction	Covariant all-probe completed transport is the remaining proof-bearing relation. It must retain response, adjoint, boundary, and holonomy while excluding off-shell nontrivial zero modes.

II The finite-prime geometry

II.1 Prime support is an exponent lattice

Definition 1 — Prime-valuation atlas and its graded fibers Let S be a finite set of primes. Its valuation lattice and positive valuation monoid are

$$V_S = \mathbb{Z}^S, \quad V_S^+ = \mathbb{N}^S.$$

Every positive integer supported on S has the unique address

$$\nu_{S(n)} = (v_{p(n)})_{p \in S}, \quad n = \prod_{p \in S} p^{v_{p(n)}}.$$

The total multiplicative degree

$$\deg(n) = |\nu_{S(n)}|_1 = \sum_{p \in S} v_{p(n)}$$

grades V_S^+ into finite simplicial fibers

$$\Delta_{S,d} = \left\{ \alpha \in \mathbb{N}^S : \sum_p \alpha_p = d \right\}.$$

A change of prime support $S \subseteq T$ supplies an inclusion of lattices and a restriction map; an analytic receiver s may specialize an address by

$$\Phi_{s(\alpha)} = \prod_{p \in S} p^{-s\alpha_p} = \exp \left(-s \sum_{p \in S} \alpha_p \log p \right).$$

Boundary. The valuation address is canonical for a declared finite prime support. A Euclidean drawing of that address, a residue chart, and the analytic specialization Φ_s are receiver faces, not additional intrinsic properties of the integer. Enlarging S changes the available atlas; it does not change the already established factorization.

The definition changes the question “Which numbers are near this prime?” into a more precise family of questions. Nearness may mean one degree-preserving root move, one residue step, one continued-fraction rebase, one analytic amplitude ratio, or one path admitted by a particular receiver. These are not interchangeable metrics.

For fixed $S = \{p_1, \dots, p_r\}$, unique factorization gives an exact coordinate chart

$$n \leftrightarrow \nu_{S(n)} = (v_{p_1}(n), \dots, v_{p_r}(n)).$$

The prime p_j is the primitive axis e_{p_j} ; the power p_j^m is the address me_{p_j} ; a semiprime $p_i p_j$ is the mixed address $e_{p_i} + e_{p_j}$. “Irreducible” is therefore a statement about the boundary of the positive monoid: a prime occupies a primitive degree-one vertex, while its later powers and products emanate along higher-degree fibers.

II.2 Exact group counts per prime axis

Theorem 2 — Exact census of a finite prime simplex If $|S| = r$, then the degree- d valuation fiber has

$$|\Delta_{S,d}| = \binom{d+r-1}{r-1}$$

members. The members using exactly k prime axes number

$$\binom{r}{k} \binom{d-1}{k-1}$$

for $1 \leq k \leq \min(r, d)$. The members containing a fixed axis p number

$$\binom{d+r-2}{r-1},$$

and those with exact exponent $v_p = m$ number

$$\binom{d-m+r-2}{r-2}$$

whenever $0 \leq m \leq d$ and $r \geq 2$.

Demonstration. The first count is the number of weak compositions of d into r coordinates. For support exactly k , choose its axes and then count the positive compositions of d into k parts. Requiring a fixed axis subtracts one from that coordinate before applying weak compositions. Fixing its exponent leaves a weak composition of $d - m$ over the remaining $r - 1$ axes.

Boundary. These counts grade multiplicative formulations by valuation degree. They do not count all integer-producing algorithms, all analytic formulations of a constant, or all receiver-equivalent paths. Such paths require additional vertical transition data.

This theorem supplies the group count previously missing from the prime observations. With three active axes,

$$|\Delta_{S,1}| = 3, \quad |\Delta_{S,2}| = 6, \quad |\Delta_{S,3}| = 10.$$

The degree-three fiber has three pure cubes, six $[2, 1]$ forms, and one full-support product. These are not empirical pattern classes; they are the permutation orbits of weak compositions of 3.

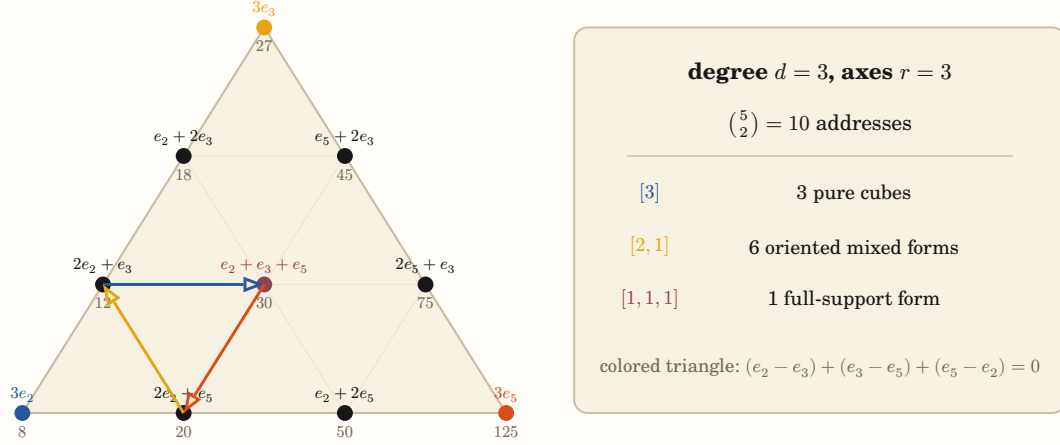


Figure 1: The degree-three valuation fiber over $S = \{2, 3, 5\}$. Its ten addresses split into the three permutation orbits $[3]$, $[2, 1]$, and $[1, 1, 1]$. The colored three-step transfer closes in the fixed chart; this is the smallest exact prime-axis A_2 loop.

For general degree d , the symmetric group on the active prime axes partitions the simplex by integer partitions of d having at most r parts. Thus

$$d = 2 : \quad [2], [1, 1],$$

and

$$d = 4 : \quad [4], [3, 1], [2, 2], [2, 1, 1], [1, 1, 1, 1].$$

The orbit label records exponent shape while the chosen axes record its address. This is one precise sense in which prime-related shapes recur self-similarly without making every instantiation identical.

II.3 The measured finite aperture

The bounded prime-spectral observation through 313 exposed 82 prime-power gears. The per-axis multiplicities were:

Prime axes	Powers admitted	Exact reason
2	8	$2^1, \dots, 2^8 \leq 313 < 2^9$
3	5	$3^1, \dots, 3^5 \leq 313 < 3^6$
5	3	$5^1, 5^2, 5^3 \leq 313 < 5^4$
7, 11, 13, 17	2 each	$p^2 \leq 313 < p^3$
58 other primes ≤ 313	1 each	$p \leq 313 < p^2$

For a cutoff X , the exact aperture on one axis is

$$a_{X(p)} = \max\{m : p^m \leq X\} = \lfloor \log_p X \rfloor.$$

Summing $a_{X(p)}$ counts prime powers without weight. Weighting each occurrence by $\log p$ gives the Chebyshev current

$$\psi(X) = \sum_{p^m \leq X} \log p.$$

The earlier scalar “82” is therefore only the cardinality of a complete path population. The atlas keeps which axis repeated, how many times it repeated, and which logarithmic displacement it contributes.

II.4 Residue loops are a different fiber

The valuation chart is multiplicative. A residue chart about a prime axis p has different exact populations:

$$\left| \frac{\mathbb{Z}}{p} \right| = p, \quad \left| \left(\frac{\mathbb{Z}}{p} \right)^\times \right| = p - 1, \quad \left| \left(\widehat{\frac{\mathbb{Z}}{p}} \right)^\times \right| = p - 1.$$

Adding a new prime p to a primordial wheel multiplies the surviving unit population by $p - 1$. The measured wheel sizes

$$1, 2, 8, 48, 480$$

for moduli

$$2, 6, 30, 210, 2310$$

are exactly Euler's totient values. They count compatible residue positions, not the exponent addresses in $\Delta_{S,d}$. The atlas relates these fibers through declared reduction maps instead of collapsing them into one notion of "prime geometry."

III Horizontal transport: roots, fractions, and recurrence

III.1 Degree-preserving prime moves

Lemma 3 — Prime-axis transfers form the root lattice A_{r-1} Enumerate $S = \{p_1, \dots, p_r\}$ and let e_p be the corresponding standard valuation axes. Every degree-preserving integral displacement is in

$$A_{r-1} = \left\{ \beta \in \mathbb{Z}^S : \sum_p \beta_p = 0 \right\},$$

and this lattice is generated by the oriented transfers

$$\alpha_{pq} = e_p - e_q.$$

In particular every oriented prime triangle closes:

$$\alpha_{pq} + \alpha_{qr} + \alpha_{rp} = 0.$$

Under the analytic specialization Φ_s , a legal transfer from q to p multiplies the observed amplitude by

$$\frac{\Phi_s(\alpha + e_p - e_q)}{\Phi_s(\alpha)} = \left(\frac{q}{p} \right)^s.$$

Demonstration. If $\beta \in \mathbb{Z}^S$ has coordinate sum zero, select one base axis p_0 . Then

$$\beta = \sum_{p \neq p_0} \beta_p (e_p - e_{p_0}),$$

because the coefficient of e_{p_0} is $-\sum_{p \neq p_0} \beta_p = \beta_{p_0}$. The triangular identity telescopes. The specialization formula follows by dividing the two Euler monomials.

Boundary. The closed triangle is flat only inside one fixed valuation and analytic chart. If the receiver, branch, completion, or admitted prime support changes along the sides, the transported triangle may carry a residual; that residual must be calculated from the transition laws and may not be inferred from the planar drawing.

The transfer $e_p - e_q$ removes one unit of exponent from the q axis and adds one to the p axis whenever the source address permits it. In a fixed degree simplex these moves traverse the adjacency graph. A fraction $\frac{p}{q}$ is therefore not an illicit float; it is an oriented valuation edge with address $e_p - e_q$.

Under

$$\Phi_{s(\alpha)} = \prod_p p^{-s\alpha_p},$$

that exact combinatorial move has analytic face

$$\Phi_{s(\alpha+e_p-e_q)} = \left(\frac{q}{p}\right)^s \Phi_{s(\alpha)}.$$

For real $s = \sigma > 0$, the magnitude orders the move. For complex $s = \sigma + it$, the same edge also carries the phase

$$\exp(it(\log q - \log p)).$$

The “probability-like” reading is consequently receiver-relative: the address is exact, while the amplitude assigned to it depends on the contemporary analytic parameter.

III.2 Continued fractions are rebase words

A positive rational path

$$x = [a_0; a_1, \dots, a_k]$$

is an exact word in unimodular transformations

$$z \mapsto a + \frac{1}{z}.$$

Its convergents are related by matrices in $\text{SL}(2, \mathbb{Z})$ up to the ordinary orientation convention. Thus a rational $\frac{p}{q}$ carries at least two complementary faces:

- its valuation displacement $\nu(p) - \nu(q)$, and
- its ordered Euclidean-algorithm word.

One records multiplicative support; the other records a causal rebase procedure. Their returned rational agrees, but neither path reconstructs the other without additional information.

III.3 The flat triangle and the possible residual

Inside one valuation chart,

$$(e_p - e_q) + (e_q - e_r) + (e_r - e_p) = 0.$$

After analytic specialization with one fixed s , the product of the three ratios is also one:

$$\left(\frac{q}{p}\right)^s \cdot \left(\frac{r}{q}\right)^s \cdot \left(\frac{p}{r}\right)^s = 1.$$

This is an exactly commuting triangular cell. Curvature does not arise by declaring the planar triangle “really higher-dimensional.” It arises when the transport changes a branch, receiver, completion, support, or boundary condition along the loop. The resulting holonomy must then be computed from those transition laws.

IV Vertical transport: formulations of π and e

IV.1 A formulation is a path, not merely its limit

Bailey’s catalogue deliberately places polygonal iterations, arctangent identities, infinite series, integrals, AGM algorithms, and digit-extraction formulas side by side [1]. MathWorld records an even wider family [2]. Their common value does not make their interiors identical:

Face	Native recurrence	Prime-sensitive interior
Polygonal	angle doubling and nested radicals	powers of 2 and chosen polygon order
Machin-like	Gaussian multiplication and argument addition	norm factorizations and branch quadrant
Euler product / L -value	multiplicative local factors	prime residue characters
Ramanujan–Chudnovsky	hypergeometric recurrence	factorial valuations and modular parameters
BBP-like	base-specific digit extraction	radix powers and residue denominators
AGM	quadratic mean iteration	duplication, elliptic/modular structure

The atlas does not need to assert that all such algorithms are one formula. It asks for explicit vertical transitions where they exist and records an open seam where they do not.

IV.2 A complete π cell: Machin through Gaussian factorization

The exact Gaussian identity

$$(5 + i)^4 = 2(1 + i)(239 + i)$$

can be verified in $\mathbb{Z}[i]$:

$$(5 + i)^4 = 476 + 480i$$

and

$$2(1 + i)(239 + i) = 476 + 480i.$$

Taking arguments in the relevant first-quadrant branches gives

$$4 \arctan\left(\frac{1}{5}\right) - \arctan\left(\frac{1}{239}\right) = \frac{\pi}{4}.$$

Taking Gaussian norms instead gives

$$N(5 + i) = 26 = 2 \cdot 13, \quad N(239 + i) = 57122 = 2 \cdot 13^4.$$

The same identity therefore has an angle face and an exact finite-prime interior on axes 2 and 13. The appearance of those primes is not a numerological scan of decimal digits; it is forced by factorization of the Gaussian integers which carry the argument relation.

IV.3 A second π cell: a character field over odd primes

Let χ_4 be the nontrivial character modulo 4. For odd primes,

$$\chi_4(p) = (-1)^{\frac{p-1}{2}} = \sin\left(\pi \frac{p}{2}\right).$$

Since $\chi_4(p)^2 = 1$,

$$1 + \chi_4 \frac{p}{p} = \frac{1 - p^{-2}}{1 - \chi_4 \frac{p}{p}}.$$

Multiplying over odd primes gives

$$\prod_{p>2} \left(1 + \chi_4 \frac{p}{p}\right) = \frac{2}{\pi}.$$

Equivalently,

$$\pi = 3 \frac{\zeta(2)}{2L(1, \chi_4)}.$$

This cell relates π to the distribution of prime axes by residue orientation. Semiprimes retain a compositional sign:

$$\chi_4(pq) = \chi_4(p)\chi_4(q).$$

The semiprime is not noise between primes; it is the product path through two local character responses.

IV.4 The e family carries factorial prime currents

The standard faces of e include the series, compound limit, continued fraction, differential flow, and exponential map [3]:

$$e = \sum_{n \geq 0} \frac{1}{n!}, \quad e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n,$$

$$e = [2; 1, 2, 1, 1, 4, 1, 1, 6, \dots], \quad y' = y, y(0) = 1.$$

For the series current $a_n = \frac{1}{n!}$,

$$a_{n+1} = \frac{a_n}{n+1}.$$

The new denominator does not enter as one opaque integer. Its prime-axis address is $\nu(n+1)$, and the accumulated denominator obeys Legendre's exact law

$$v_{p(n!)} = \sum_{k \geq 1} \left\lfloor \frac{n}{p^k} \right\rfloor.$$

Thus the factorial series traverses every prime axis when its powers occur in the successor sequence. Its return toward e compresses those paths into a contemporary approximation without making their recurrence unavailable to a receiver which retains the series.

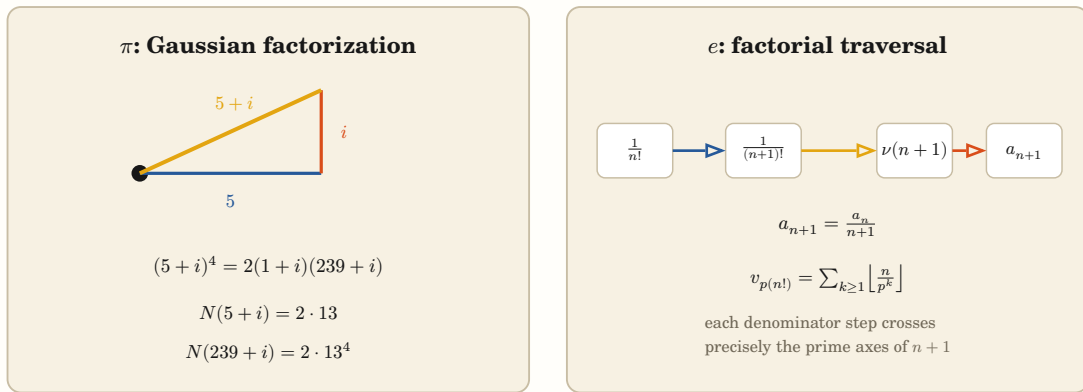


Figure 2: Two constant-formulation cells expose different prime-axis interiors. Machin's arctangent identity closes through exact Gaussian factorization; the factorial series for e walks through the valuation address of every successor denominator. The constant is the shared return, not the loss of the paths which produced it.

IV.5 A series has a term current and an accumulation current

The preceding examples name prime factors inside formulas. The next question is stricter: how do those factors move while the formula is actually evaluated?

For the arctangent family

$$A\left(\frac{u}{v}\right) = \sum_{n \geq 0} (-1)^n \frac{u^{2n+1}}{v^{2n+1}(2n+1)},$$

put

$$a_n = (-1)^n \frac{u^{2n+1}}{v^{2n+1}(2n+1)}.$$

Then

$$r_n = \frac{a_{n+1}}{a_n} = -u^2 \frac{2n+1}{v^2(2n+3)}.$$

If $D_n = \text{div}(a_n)$ is its signed rational prime divisor, the transition is

$$D_{n+1} - D_n = 2 \text{div}(u) - 2 \text{div}(v) + \text{div}(2n+1) - \text{div}(2n+3).$$

The odd factor is therefore a moving carrier. The current face $2n+1$ returns on the numerator hand and departs while $2n+3$ enters on the denominator hand. The parameter v^2 supplies a persistent rail. These two currents collide at the exact modular sheets

$$2n+1 = 0 \bmod p^k \quad \text{or} \quad 2n+3 = 0 \bmod p^k,$$

for p^k dividing v . Since 2 is invertible modulo every odd p^k , each condition selects one residue class. This is a parameter-relative phase relation inside one recurrence species.

Term formation is still only one face. With

$$S_n = \sum_{j=0}^n a_j,$$

the complete first-person transition is

$$\begin{pmatrix} a_{n+1} \\ S_{n+1} \end{pmatrix} = \begin{pmatrix} r_n & 0 \\ r_n & 1 \end{pmatrix} \begin{pmatrix} a_n \\ S_n \end{pmatrix}.$$

For reduced $S = \frac{A}{B}$ and $a = \frac{C}{D}$, the accumulated successor is

$$\frac{A}{B} + \frac{C}{D} = \frac{AD + BC}{BD} \xrightarrow{g=\text{gcd}(A \cdot D + B \cdot C, B \cdot D)} \frac{\frac{AD+BC}{g}}{B \frac{D}{g}}.$$

This gcd quotient is an exact outgoing compression event. It removes denominator directions which no longer factor into the sum while retaining the reduced consequence. It is distinct from cancellation between the numerator and denominator channels of r_n .

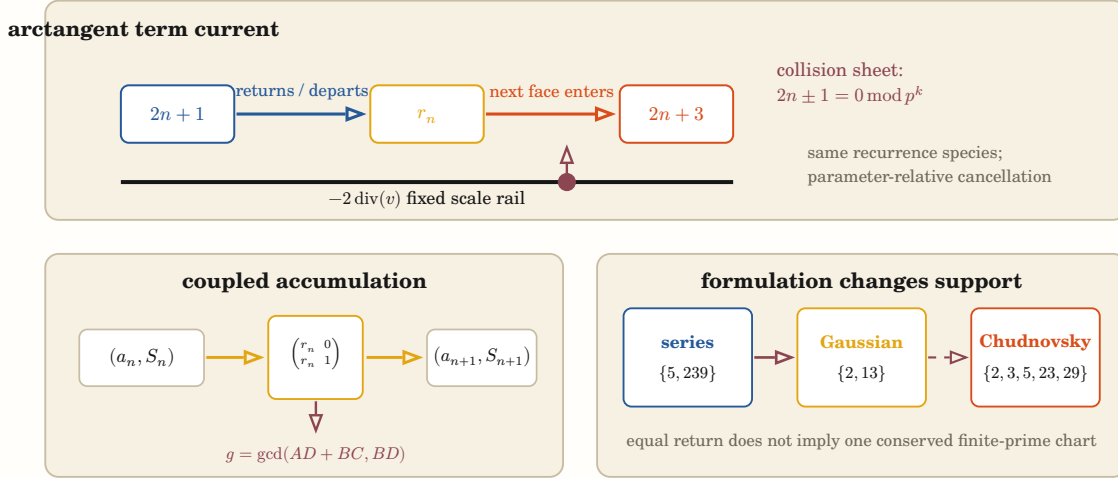


Figure 3: The measured presentation topology. The arctangent recurrence carries a moving odd face over a fixed parameter rail; modular collisions alter cancellation without changing the formula species. Partial sums form a second coupled topology through an exact gcd quotient. Vertical formulation transport can change finite-prime support, so the common transcendental return cannot replace its paths.

IV.5.1 Exact bounded presentation-topology cut

The accompanying instrument carries 12 exact terms of $A(1)$, $A(\frac{1}{5})$, and $A(\frac{1}{239})$, plus eight Chudnovsky summands. Each event retains named factor channels before reduction, both prime-valuation hands, internal annihilation, the surviving signed divisor, the coupled term/sum matrix, the additive gcd, and the next denominator divisor. The complete paths are deposited at observations/eros-transcendental-presentation-topology-01/REPORT.json.

At the twelfth arctangent term the divisors are

$$\begin{aligned} \text{div}(a_{11}(1)) &= -[23], \\ \text{div}\left(a_{11}\left(\frac{1}{5}\right)\right) &= -23[5] - [23], \quad \text{div}\left(a_{11}\left(\frac{1}{239}\right)\right) = -23[239] - [23]. \end{aligned}$$

All preceding odd faces have departed from the current term. The received partial sums retain different denominator topology:

Path	Internal recurrence cancellation	Final partial denominator
$A(1)$	none	$3^2 \cdot 5 \cdot 7 \cdot 11 \cdot 13 \cdot 17 \cdot 19 \cdot 23$
$A(\frac{1}{5})$	5^2	$3^2 \cdot 5^{23} \cdot 7 \cdot 11 \cdot 13 \cdot 17 \cdot 19 \cdot 23$
$A(\frac{1}{239})$	none	$3^2 \cdot 5 \cdot 7 \cdot 11 \cdot 13 \cdot 17 \cdot 19 \cdot 23 \cdot 239^{23}$

The $\frac{1}{5}$ arm crosses two fixed/moving 5-axis collisions in this aperture; the $\frac{1}{239}$ arm has not yet reached a moving odd multiple of 239. This is why structurally identical recurrences have nonidentical local annihilation.

Semiprime shapes are likewise local. The fixed channel 5^2 is a $[2]$ face at every $A(\frac{1}{5})$ transition, while an odd factor such as $15 = 3 \cdot 5$ is a $[1, 1]$ face. A later channel coupling may remove part of either. “Semiprime” therefore addresses a channel, hand, and event grain; it is not a label on the transcendental return.

IV.5.2 Chudnovsky is the same carrier law at higher arity

For

$$c_n = (-1)^n (6n)! \frac{A + Bn}{(3n)! (n!)^3 640320^{3n}},$$

where $A = 13591409$ and $B = 545140134$,

$$\frac{c_{n+1}}{c_n} = - \frac{\prod_{j=1}^6 (6n + j)}{\left(\prod_{j=1}^3 (3n + j) \right) (n+1)^3 640320^3} \cdot \frac{A + B(n+1)}{A + Bn}.$$

The event is not one opaque hypergeometric coefficient. It couples a sixfold factorial lift, a threefold lift, a cubic successor, the fixed scale

$$640320^3 = (2^6 \cdot 3 \cdot 5 \cdot 23 \cdot 29)^3,$$

and a moving linear quotient. Each $A + Bn$ enters as a `linear_successor`, becomes the following event's `linear_predecessor`, and then departs. The initial face is already a mixed semiprime:

$$13591409 = 13 \cdot 1045493.$$

Across eight terms, the exact internal recurrence cancellation is

$$2^{39} \cdot 3^{19} \cdot 5^{10} \cdot 7^3 \cdot 11 \cdot 13 \cdot 17 \cdot 19 \cdot 23 \cdot 29,$$

while additive accumulation removes repeated fixed-scale directions and returns the finite partial denominator

$$2^{117} \cdot 3^{17} \cdot 5^{19} \cdot 23^{20} \cdot 29^{20}.$$

These exponent populations count actual event-local cancellation; they are not a retained global mass.

IV.5.3 One return, nonidentical finite-prime charts

The two Machin partials remain a vector until the exact map $[4, -1]$ is applied. Their alternating tails retain an exact rational residual interval. The completed argument relation closes through

$$(5 + i)^4 = 2(1 + i)(239 + i),$$

with oriented factorizations

$$5 + i = (1 + i)(3 - 2i), \quad 239 + i = i(1 + i)(3 - 2i)^4.$$

Consequently the local supports are:

Presentation face	Persistent scale or norm axes
Machin series	$\{5, 239\}$
Machin Gaussian proof	$\{2, 13\}$
Chudnovsky series	$\{2, 3, 5, 23, 29\}$

The Gaussian proof uses neither Machin denominator axis. Only 5 is shared between the two series-scale charts. Equal π return therefore does not conserve one hidden finite-prime support. A vertical formulation map changes support through its actual Gaussian, hypergeometric, modular, branch, and evaluation laws.

IV.6 The π -e seam is exponential covering geometry

The identity

$$\exp(i\pi) + 1 = 0$$

relates e and π through a map of geometric species. The exponential map sends the additive imaginary line to the unit circle and identifies translations by $2\pi i$. The point $i\pi$ is a half-turn lift of -1 . This is the rigorous point–line–loop relation in this cell:

- $i\pi$ is a point in the covering coordinate;
- a path from 0 to $i\pi$ is an oriented line occurrence;
- its exponential image is a half-loop on the unit circle; and
- changing the logarithm branch changes the lift by $2\pi ik$ without changing the returned point.

The same returned point therefore admits multiple lineages. Branch data is part of the atlas cell, not an error to discard.

V The exact prime–archimedean bridge

V.1 The product formula

“Prime–archimedean” is not only a metaphor. For a nonzero rational

$$x = \pm \prod_p p^{v_{p(x)}},$$

normalize the p -adic magnitudes by

$$|x|_p = p^{-v_{p(x)}}.$$

Then

$$|x|_\infty \prod_p |x|_p = 1.$$

Taking logarithms yields

$$\log |x|_\infty = \sum_p v_{p(x)} \log p.$$

The ordinary real magnitude and all finite valuation axes are exact co-faces of the same rational occurrence. Neither side is an enlarged absolute field which contains the other; their normalized product closes only when all places participating in that rational are included.

This identity supplies a first atlas square which genuinely commutes: factor the rational horizontally into prime valuations, or measure its real magnitude vertically and take logarithms. Both paths return the same weighted sum.

V.2 Situated mean transport: the continuous companion to a discrete event

Lemma 4 — Mean transport is path-relative and seams add discrete terms Let (M, g) be a Riemannian manifold, let $\gamma : [0, L] \rightarrow M$ be a unit-speed geodesic from p to q , and let $f : M \rightarrow \mathbb{R}$ be smooth on the traversed chart. Then

$$f(q) - f(p) = \int_0^L df_{\gamma(s)}(\dot{\gamma}(s)) \, ds,$$

and there is a $c \in (0, L)$ such that

$$f(q) - f(p) = L df_{\gamma(c)}(\dot{\gamma}(c)) = L g_{\gamma(c)}(\text{grad}_g f|_{\gamma(c)}, \dot{\gamma}(c)).$$

For two scalar fields P, Q , Cauchy's mean-value theorem along the same path gives a c with

$$(P(q) - P(p))dQ_{\gamma(c)}(\dot{\gamma}(c)) = (Q(q) - Q(p))dP_{\gamma(c)}(\dot{\gamma}(c)).$$

More generally, let $V(s) \in E_{\gamma(s)}$ be a section along γ of a vector bundle with connection, and let $\mathcal{P}_{s \rightarrow 0}$ denote parallel transport into the initial fiber. Then

$$\mathcal{P}_{L \rightarrow 0} V(L) - V(0) = \int_0^L \mathcal{P}_{s \rightarrow 0} \nabla_{\dot{\gamma}} V(s) ds.$$

If a piecewise-smooth parameter path crosses typed seams $s_1, \dots, s_m$, its complete scalar transport decomposes as

$$f(\gamma(L)) - f(\gamma(0)) = \sum_j \int_{I_j} df(\dot{\gamma}) ds + \sum_k \Delta_k,$$

where the Δ_k are the declared branch, rank, phase, or discrete valuation changes at those seams.

Demonstration. Apply the fundamental theorem of calculus and the ordinary scalar mean-value theorem to the pullback $f \circ \gamma$. Apply Cauchy's theorem to $P \circ \gamma$ and $Q \circ \gamma$. For the bundle statement, parallel-transport $V(s)$ into the fixed vector space E_p ; the derivative of that pulled-back curve is $\mathcal{P}_{s \rightarrow 0} \nabla_{\dot{\gamma}} V(s)$, so ordinary vector-space integration gives the identity. Split a piecewise-smooth path at every seam and telescope its endpoint changes; the unmatched one-sided values are exactly the Δ_k .

Boundary. The witness c belongs to the chosen path and need not be invariant under another route or parameterization. A vector-valued mean-value equality at one point generally does not follow: values first require a connection and parallel transport into one fiber. Transport around different paths may carry holonomy. A discrete seam may not be hidden inside a smooth derivative or inferred from an averaged scalar.

Corollary 5 — Equal return forces a situated tangency or a seam Under the smooth scalar hypotheses of the situated mean-transport lemma, if $f(q) = f(p)$, there is a $c \in (0, L)$ such that

$$df_{\gamma(c)}(\dot{\gamma}(c)) = g_{\gamma(c)}(\text{grad}_g f|_{\gamma(c)}, \dot{\gamma}(c)) = 0.$$

At a regular point of a surface or higher-dimensional manifold, the path tangent is therefore tangent to the level hypersurface $f^{-1}(f(p))$ at $\gamma(c)$. If an entire smooth family preserves the return, then

$$df_{\gamma(s)}(\dot{\gamma}(s)) = 0$$

at every regular parameter value, so its tangent current lies in $\ker(df)$ throughout the family.

For a piecewise-smooth path with declared seams and the same endpoint return, the complete closure law is instead

$$\sum_j \int_{I_j} df(\dot{\gamma}) ds = - \sum_k \Delta_k.$$

A path with no smooth tangency can therefore close only through one or more nonzero seam terms.

Demonstration. The first statement is Rolle's theorem applied to $f \circ \gamma$, together with the Riemannian identity $df(X) = g(\text{grad}_g f, X)$. If $f \circ \gamma$ is constant, its derivative vanishes everywhere. The seam statement follows by setting the endpoint difference to zero in the piecewise mean-transport identity.

Boundary. Tangency to a level set does not imply $\text{grad}_g f = 0$, identify a unique point, or make the witness independent of the path. Equal returned scalar values do not identify the paths' factor support, branch, holonomy, or interior topology. A nonsmooth, multivalued, or bundle-valued return requires its declared chart, connection, and seam data before this scalar corollary applies.

This is the precise meaning of a parameter $x = x(t)$ in the atlas. Scalar mean transport is obtained only after an admitted path has selected the tangent along which change is read; Cauchy's version compares two scalar responses on that same path [4]. For the cross-ratio swing $\chi = \frac{P}{Q}$,

$$d\chi = \frac{QdP - PdQ}{Q^2}.$$

Thus $Q = 0$ is a chart boundary and the mean witness is relative to the chosen path. Bundle-valued transport additionally requires a connection before endpoint values occupy one comparable fiber [5], [6].

Applied to presentation topology, the smooth pieces I_j carry deformation of coefficients, scale, or evaluation point. The seam terms Δ_k carry modular collisions, support changes, branch crossings, and rank loss. The theorem therefore does not smooth away quantization: it specifies the within-phase current against which each discrete change is measured. A second route may return a different parallel transport, making holonomy the residual of comparing complete paths rather than isolated endpoint values.

This gives three distinct geometries for a shared transcendental return. An exact same-return family lies tangent to the kernel of its evaluation differential at every smooth point. A path whose endpoints merely agree must have at least one situated level-set tangency. A support-changing path may instead balance its smooth transport against explicit seam terms. Those cases cannot be inferred from the endpoint value alone.

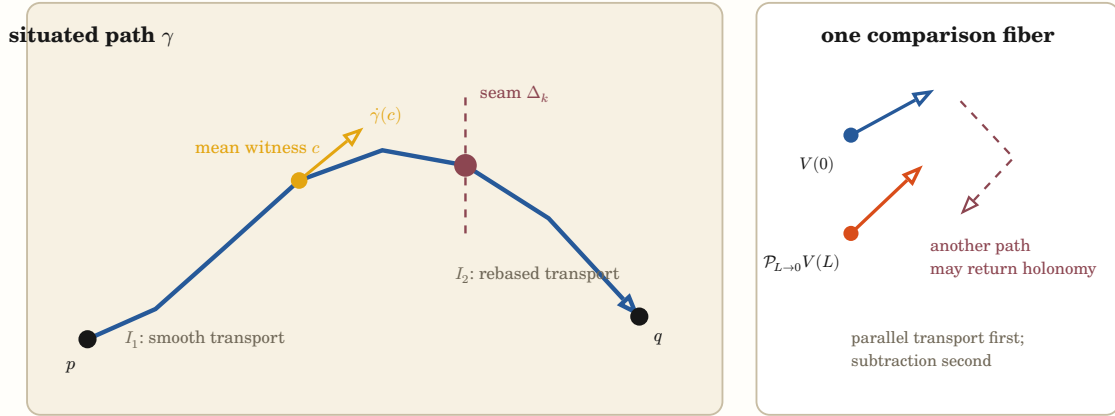


Figure 4: Mean transport on a manifold is path-relative. A scalar field admits a one-dimensional mean witness along the chosen γ . Bundle-valued states must first be parallel-transported into one fiber, and a discrete branch, rank, phase, or valuation seam contributes an explicit jump Δ_k . Curvature becomes visible as path-dependent holonomy, not as an absolute coordinate bend.

V.3 Zeta completion adds the infinite place

For $\text{Re}(s) > 1$,

$$\zeta(s) = \sum_{n \geq 1} n^{-s} = \prod_{p(1-p^{-s})}^{-1}.$$

The Euler product assembles the finite-prime receivers. The completed function

$$\Xi(s) = \frac{1}{2} s(s-1) \pi^{-\frac{s}{2}} \Gamma\left(\frac{s}{2}\right) \zeta(s)$$

also includes the archimedean factor and satisfies

$$\Xi(s) = \Xi(1-s).$$

The Gaussian

$$g(x) = e^{-\pi x^2}$$

is fixed by the standard Fourier transform normalization. Summing its lattice translates gives theta; Poisson transport supplies reciprocal scale; Mellin transport produces the Gamma factor and the functional reflection [7], [8].

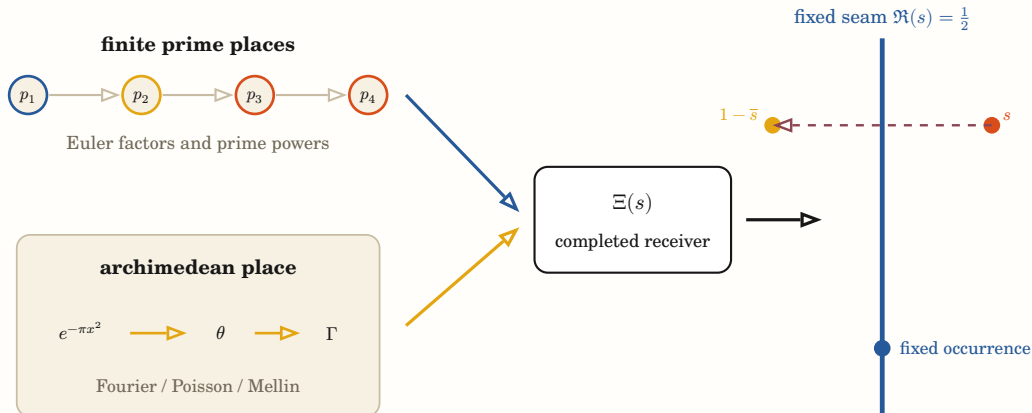


Figure 5: The completed zeta receiver joins finite prime places to the archimedean Gaussian–theta–Mellin chain. The critical line is the fixed seam of $s \mapsto 1 - \bar{s}$, not a preferred line drawn independently of completion. The atlas question is whether transported response remains sign-coherent around every admissible completed loop.

The critical line is therefore not an arbitrary Euclidean line imposed on the prime picture. After completion it is the fixed seam of the antiholomorphic return

$$J(s) = 1 - \bar{s}.$$

Centering at $s = \frac{1}{2}$ writes $s = \frac{1}{2} + z$ and turns the return into

$$z \mapsto -\bar{z}.$$

Another valid chart may curve the seam. What is invariant is the fixed locus and the completed response carried across it.

V.4 The conformal rebase atlas: one normal hand in four charts

A transformed grid is useful only after its crossings are typed. Let the first nonzero local term of a holomorphic map at z_0 be

$$F(z_0 + \delta) = F(z_0) + c_m \delta^m + O(\delta^{m+1}), \quad c_m \neq 0.$$

The pair $(F(z_0), m)$ separates four events which a flattened image can make look alike:

Local event	Exact condition	Carried geometry
Regular crossing	$F(z_0) \neq 0, m = 1$	$F'(z_0)$ transports both source axes; their angle is preserved.
Simple zero	$F(z_0) = 0, m = 1$	Radius collapses, phase winds once, and the local tangent remains regular.
Critical branch	$F(z_0) \neq 0, m > 1$	The first derivative vanishes; local degree, rather than a fabricated tangent, determines the branch rays.
Multiple zero	$F(z_0) = 0, m > 1$	Zero order m , critical order $m - 1$, and phase winding m coexist but remain separately named.

There is a fifth visual possibility which is not a fifth germ species: distinct points $z_1 \neq z_2$ may satisfy $F(z_1) = F(z_2)$. Their source lineages and derivative hands must remain plural. The bounded exact cell $F(z) = z^2$ retains sources $+1$ and -1 , common output 1 , and derivative hands $+2$ and -2 . Equal output placement therefore does not identify the paths which reached it.

Away from zeros and critical points, write

$$U = \log|F|, \quad V = \arg F.$$

The Cauchy–Riemann relations make (U, V) an orthogonal potential net:

$$\nabla U \cdot \nabla V = 0, \quad |\nabla U|^2 = |\nabla V|^2.$$

The measured affine germ carries this without approximation:

$$\nabla U = \left(\frac{399}{205}, -\frac{1008}{205} \right), \quad \nabla V = \left(\frac{1008}{205}, \frac{399}{205} \right).$$

At a zero of order m , U has the logarithmic singularity

$$U = \log|c_m| + m \log|\delta| + \dots$$

while V winds m full turns. At a nonzero critical point of local degree m , the first nonconstant term of $\log F$ is proportional to δ^m ; each real or imaginary level family has $2m$ incident rays. A zero, a critical point, and an intersection are consequently related but not interchangeable classifications.

Now factor the completed-zeta occurrence itself. Besides the holomorphic functional reflection

$$R(s) = 1 - s, \quad \Xi(R(s)) = \Xi(s),$$

retain the antiholomorphic hand

$$J(s) = 1 - \bar{s}, \quad \Xi(J(s)) = \overline{\Xi(s)}.$$

Its exact orbit projectors are

$$P_{\text{fix}(s)} = \frac{s + J(s)}{2} = \frac{1}{2} + i \operatorname{Im}(s),$$

$$P_{\text{normal}(s)} = \frac{s - J(s)}{2} = \operatorname{Re}(s) - \frac{1}{2}.$$

Thus every s is reconstructed as $P_{\text{fix}(s)} + P_{\text{normal}(s)}$. The critical value $\frac{1}{2}$ is supplied by the fixed projector of this order-two orbit. It is not an absolute declaration that one side is positive, nor a mean-value witness chosen on a plotted interval. Writing

$$s = \frac{1}{2} + \varepsilon + it$$

makes ε the signed normal hand. On $\text{Fix}(J)$ a response pairs with itself as a squared magnitude. Off the fixed seam its partner is a cross-pair, so positivity cannot be assigned merely by turning the picture.

The same ε enters logarithmic scale. For

$$u = \log\left(\frac{x}{x_0}\right),$$

carry the rebase character as the exact word

$$\rho_{\varepsilon,t}(u) = \exp(\varepsilon u) \exp(itu).$$

Its primary coordinates are $(u, \varepsilon u, tu)$: source scale, log-radius, and phase. No trigonometric or exponential decimal is required. $\varepsilon = 0$ gives a constant-radius orbit. The two functional partners $+\varepsilon$ and $-\varepsilon$ have identical phase and reciprocal radius at every u , producing the two opposed logarithmic spirals.

The centered Smith receiver turns this same hand into shell incidence. For $w = \varepsilon + it$ and rational $a > 0$, set

$$W_{a(w)} = \frac{w - a}{w + a}.$$

Direct ratio algebra gives

$$|W_{a(w)}|^2 - 1 = \frac{-4a\varepsilon}{|w + a|^2}.$$

Hence

$$\varepsilon = 0 \quad \text{iff} \quad w \in \text{Fix}(J) \quad \text{iff} \quad \rho_{\varepsilon, t} \text{ has constant radius} \quad \text{iff} \quad |W_{a(w)}| = 1.$$

The sphere lift

$$S(W) = \left(\frac{2 \operatorname{Re} W}{1 + |W|^2}, \frac{2 \operatorname{Im} W}{1 + |W|^2}, \frac{|W|^2 - 1}{1 + |W|^2} \right)$$

exposes the oriented normal without changing the tangential point. At $t = \frac{3}{2}$, $a = 1$, and $\varepsilon = \pm \frac{1}{4}$, the exact pair is

$$\begin{aligned} W_1 \left(\frac{1}{4} + 3\frac{i}{2} \right) &= \frac{21}{61} + 48\frac{i}{61}, \quad |W_1|^2 = \frac{45}{61}, \\ W_1 \left(-\frac{1}{4} + 3\frac{i}{2} \right) &= \frac{7}{15} + 16\frac{i}{15}, \quad |W_1|^2 = \frac{61}{45}. \end{aligned}$$

Their squared magnitudes multiply to one. On the sphere they become

$$\left(\frac{21}{53}, \frac{48}{53}, -\frac{8}{53} \right), \quad \left(\frac{21}{53}, \frac{48}{53}, +\frac{8}{53} \right).$$

The tangential coordinates agree while the normal hands oppose exactly. On the seam, $w = 3\frac{i}{2}$ and $w = 2i$ instead give the rational shell points $(\frac{5}{13}, \frac{12}{13}, 0)$ and $(\frac{3}{5}, \frac{4}{5}, 0)$.

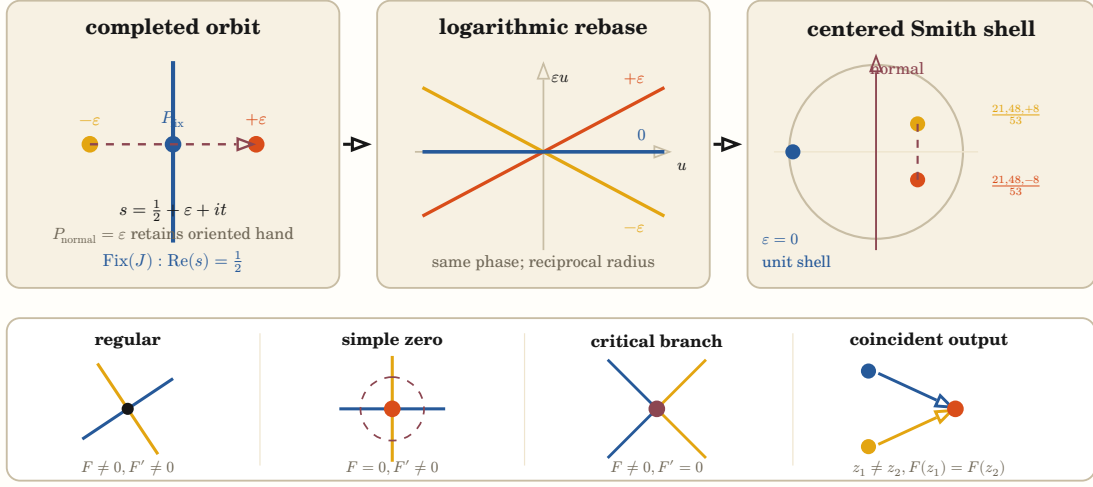


Figure 6: The exact conformal-rebase carrier. The completed involution first factors an occurrence into fixed and signed-normal parts. The same normal coordinate becomes the real exponent of logarithmic rebase and the signed normal of the centered Smith sphere; its two nonzero hands retain reciprocal radius. The lower strip keeps four local events distinct before any rendering: regular crossing, simple zero, critical branch, and distinct-source coincidence.

This also specifies a lawful revolving-zeta chart. For a declared source line $s = \sigma + it$, use

$$C_\sigma^{F(t)} = (t, \text{Re } F(\sigma + it), \text{Im } F(\sigma + it)).$$

The retained t is source chronology, not an outside camera coordinate. $F = 0$ is an axis hit; $F' = 0$ is a critical branch. They coincide only at a multiple zero. Two source values meeting one rendered point remain two lineages. Revolution is licensed as a constant-radius shell only when the actual rebase has $\varepsilon = 0$; otherwise the lawful figure is a spiral.

Raw zeta and completed zeta must also remain different chart faces. For example, $\zeta(-2) = 0$ is a simple trivial zero, yet the Gamma pole cancels it under completion and

$$\Xi(-2) = \Xi(3) = 3 \frac{\zeta(3)}{2\pi}.$$

A raw-zeta axis hit is therefore not automatically a completed- Ξ axis hit. The exact appearances of π in the completion are joined to the retained series and Gaussian-factorization carriers developed earlier in this paper; they are not evaluated as floating constants.

V.5 The reciprocal chord acquires a discrete connection

The preceding sphere pair is not exhausted by its two plotted endpoints. Carry one oriented chord at event k as

$$q_k \in \mathbb{Q}^2, \quad u_k \in S^2(\mathbb{Q}), \quad P_{\pm,k} = q_k \pm r u_k.$$

Here q_k is the chord's actual intersection with the declared complex plane, not an outside-camera position. One exact discrete transport is

$$q_{k+1} = q_k + \Delta q_k, \quad u_{k+1} = U_k u_k, \quad \Delta P_{\pm} = \Delta q_k \pm r \Delta u_k.$$

For the reciprocal Smith chord at $t = \frac{3}{2}$, $a = 1$, and $\varepsilon = \pm \frac{1}{4}$, an exact rational tilt sends the vertical axis to

$$u_1 = \left(\frac{7}{25}, 0, \frac{24}{25} \right)$$

and moves its plane intersection to

$$q_1 = \left(\frac{175}{424}, \frac{48}{53}, 0 \right).$$

The plane cut is no longer the chord midpoint: its two pole distances are $\frac{15}{424}$ and $\frac{113}{424}$, while their sum remains the fixed chord length $\frac{16}{53}$. The observer cut has changed; the rigid chord has not.

A following precession around the plane normal carries the exact phase

$$\varphi = \frac{3}{5} + 4\frac{i}{5}.$$

It gives

$$u_2 = \left(\frac{21}{125}, \frac{28}{125}, \frac{24}{25} \right), \quad q_2 = -\frac{1011}{2120} + 463\frac{i}{530},$$

and, for the stereographic direction coordinate ζ ,

$$q_2 = \varphi q_1, \quad \zeta(u_2) = \varphi \zeta(u_1), \quad \zeta(u_1) = 7.$$

Every pole remains exactly on the unit sphere and the chord's squared separation remains $\frac{256}{2809}$. The two spherical pole arcs are retained symbolically, rather than numerically collapsed, by

$$\ell_+ = \arccos\left(\frac{210837}{351125}\right), \quad \ell_- = \arccos\left(\frac{5496717}{8778125}\right).$$

The two pole paths factor common motion from oriented coupling:

$$S = \|\Delta P_+\|^2 + \|\Delta P_-\|^2 = 2\|\Delta q\|^2 + 2r^2\|\Delta u\|^2,$$

$$D = \|\Delta P_+\|^2 - \|\Delta P_-\|^2 = 4r\Delta q \cdot \Delta u, \quad \chi = \frac{D}{S}.$$

Pure precession gives the pair $(\frac{18}{25}, \frac{18}{25})$; translation orthogonal to the precession gives $(\frac{68}{25}, \frac{68}{25})$. Coupled translation and precession instead give

$$\left(\|\Delta P_+\|^2, \|\Delta P_-\|^2 \right) = \left(\frac{13}{25}, \frac{73}{25} \right), \quad \chi = -\frac{30}{43}.$$

Pole exchange preserves S and reverses χ to $\frac{30}{43}$. Thus χ is an oriented edge proportion, not an intrinsic scalar state of either pole.

For parameters λ^a , the two pole sheets induce

$$g_{ab}^\pm = \partial_a P_\pm \cdot \partial_b P_\pm.$$

The exact two-parameter cell returns

$$g^+ = \begin{pmatrix} \frac{5}{4} & \frac{5}{6} \\ \frac{5}{6} & \frac{10}{9} \end{pmatrix}, \quad g^- = \begin{pmatrix} \frac{5}{4} & -\frac{5}{6} \\ -\frac{5}{6} & \frac{10}{9} \end{pmatrix}.$$

Both determinants are $\frac{25}{36}$. Their average retains common activity,

$$g^{\text{common}} = \begin{pmatrix} \frac{5}{4} & 0 \\ 0 & \frac{10}{9} \end{pmatrix},$$

while their half-difference retains the hand,

$$g^{\text{coupling}} = \begin{pmatrix} 0 & \frac{5}{6} \\ \frac{5}{6} & 0 \end{pmatrix}.$$

Averaging the sheets alone would erase exactly the translation–precession coupling which makes their paths nonidentical.

Finally, a triangular face compares direct transport U_{02} with composed transport $U_{12}U_{01}$ through

$$H_{012} = U_{02}^{-1}U_{12}U_{01}.$$

The flat foil returns $H_{012} = I$. The mixed-axis foil returns

$$H_{012} = \begin{pmatrix} 0 & 0 & -1 \\ 0 & -1 & 0 \\ -1 & 0 & 0 \end{pmatrix}$$

and direct-versus-composed pole discrepancy $\frac{98}{25}$. This is discrete connection curvature: two routes reach one base vertex with different local frames.

The independent conic pencil

$$Q_{\mu(x,y)} = x^2 + \mu y^2 - 1$$

has determinant $-\mu$. Ellipse, circle, and hyperbola are real affine faces of a nonsingular projective conic; at $\mu = 0$ the conic becomes the line pair $(x - 1)(x + 1) = 0$. The discriminant is therefore a typed phase seam, but the construction does not assert that it causes the chord precession or the face holonomy.

This closes a carrier, not the RH implication. The rotations in the cell are controlled foils. A proof-bearing construction must derive the transport U_k , the relevant sheet metric, and its response from the completed Ξ field itself, then show that every genuine nontrivial zero mode has vanishing normal chord.

VI The formulation atlas as a bicategory with an exact rechart core

VI.1 Objects, arrows, and two-cells

Definition 6 — Situated formulation family Let $\mathcal{C}$ be a category of admissible spaces and maps, let B be a parameter base, and work in the slice $\frac{\mathcal{C}}{B}$. A **situated formulation family** over B is a decorated object

$$\mathcal{F}_B = (u_{\mathcal{F}} : U_{\mathcal{F}} \rightarrow B, L_{\mathcal{F}}, \Delta_{\mathcal{F}}, \mathcal{R}_{\mathcal{F}}, c_{\mathcal{F}}, \pi_{\mathcal{F}}).$$

The total space $U_{\mathcal{F}}$ retains the parameters and internal states admitted by the typed formulation law $L_{\mathcal{F}}$. The typed discriminant $\Delta_{\mathcal{F}}$ records where a required branch, rank, convergence, operation, or receiver law fails to continue. The receiver family

$$\mathcal{R}_{\mathcal{F}} = \{q_{\rho, \mathcal{F}} : U_{\mathcal{F}} \rightarrow Y_{\rho}\}_{\rho}$$

presents selected faces. The optional cost or metric record $c_{\mathcal{F}}$ and source testimony $\pi_{\mathcal{F}}$ are decorations, not the identity of the family.

Boundary. A specialization must name $\mathcal{C}$, the parameter base, the complete domain of each law, and every receiver used in a comparison. Equality after one receiver does not identify formulation families, their internal states, or their discriminants.

Definition 7 — Situated formulation-span atlas and exact rechart core Assume $\mathcal{C}$ has pullbacks, fix a parameter base B , and choose formulation decorations stable under identity spans and

pullback composition. A **formulation correspondence** from $\mathcal{F}_B$ to $\mathcal{G}_B$ is an admissibly decorated span in $\frac{\mathcal{C}}{B}$,

$$U_{\mathcal{F}} \xleftarrow{\ell} W \xrightarrow{r} U_{\mathcal{G}}, \quad u_{\mathcal{F}} \circ \ell = u_{\mathcal{G}} \circ r.$$

Its apex W is the supplied overlap or joint construction; its decorations state which laws, boundaries, and receiver comparisons are actually transported. It is **exact** for a declared receiver subfamily $\mathcal{R}_0$ when

$$q_{\rho, \mathcal{F}} \circ \ell = q_{\rho, \mathcal{G}} \circ r \quad \text{for every } \rho \in \mathcal{R}_0.$$

A 2-cell between parallel correspondences is a map of their apexes which commutes with both legs and preserves the declared decorations.

These objects, correspondences, and 2-cells form the **situated formulation-span atlas** FormSpan_B . Its maximal sub-bicategory containing only equivalence correspondences and invertible 2-cells is the **exact rechart bigroupoid**. Identifying invertible 2-isomorphic arrows gives an ordinary rechart groupoid. A map of parameter bases $f : B' \rightarrow B$ acts by pullback wherever every required decoration also pulls back lawfully.

Boundary. Pullbacks of bare spans do not by themselves prove that a chosen decoration class is closed; that stability is a hypothesis of the atlas. A residual between receiver values is testimony about a failed comparison; it is not automatically a 2-cell. A one-way limit, quotient, projection, branch restriction, or compression may be a lawful correspondence without belonging to the rechart core. Base change gives an indexed bicategory; it is not called a stack without descent data and descent proofs.

Lemma 8 — Formulation correspondences compose by shared interior Assume the chosen formulation decorations are stable under identity spans and pullback composition. Let

$$U_{\mathcal{F}} \leftarrow W_{\mathcal{F}\mathcal{G}} \rightarrow U_{\mathcal{G}} \quad \text{and} \quad U_{\mathcal{G}} \leftarrow W_{\mathcal{G}\mathcal{H}} \rightarrow U_{\mathcal{H}}$$

be formulation correspondences over B . Their composite has apex

$$W_{\mathcal{F}\mathcal{G}} \times_{U_{\mathcal{G}}} W_{\mathcal{G}\mathcal{H}}.$$

The diagonal span is an identity, and composition is associative up to the canonical pullback isomorphism. If both input correspondences are exact for the same receiver family $\mathcal{R}_0$, then their composite is exact for $\mathcal{R}_0$.

Demonstration. The pullback consists precisely of pairs of supplied comparison states whose two middle legs name the same $\mathcal{G}$ -state. Its projections followed by the outer legs give a span from $\mathcal{F}$ to $\mathcal{H}$. Pullback uniqueness supplies the unitors and associator. For each $\rho \in \mathcal{R}_0$, exactness identifies the $\mathcal{F}$ and $\mathcal{G}$ receiver faces on the first apex and the $\mathcal{G}$ and $\mathcal{H}$ faces on the second. Equality is therefore transitive on the pullback.

Boundary. The pullback matches only the middle data explicitly retained by the spans. It does not recover directions already discarded by either leg, and it does not turn a noninvertible correspondence into a rechart. Decoration stability must be proved for each specialization.

The prime–archimedean specialization of one family may be written schematically as

$$O = (S, \alpha, F, R, B),$$

where:

- S is its admitted finite-prime support;
- $\alpha \in \mathbb{Z}^S$ is its valuation address or addressed population;
- F is the formulation law and parameter path;
- R is the receiver or chart through which it is evaluated; and
- B is the declared boundary data, including branch, convergence, orientation, and completion.

A live parametric occurrence is a trajectory

$$\gamma(t) = (S(t), \alpha(t), F_t, R_t, B_t).$$

On a regular interval, S and the boundary type may remain fixed while the formulation and receiver deform continuously. A new prime support, branch change, singular rank, or period failure is a discrete seam at which the trajectory enters another chart. Parameterization therefore changes the speed and order of traversal without turning distinct seams into one absolute timeline.

Horizontal correspondences change prime support or valuation address. Vertical correspondences change formulation or receiver. Their apex retains the actual joint interior on which the comparison is defined. An invertible local rechart belongs to the exact rechart core; a one-way limit, quotient, root selection, bare summation, or projection remains a lawful span outside that core and must keep its lost directions in the boundary record.

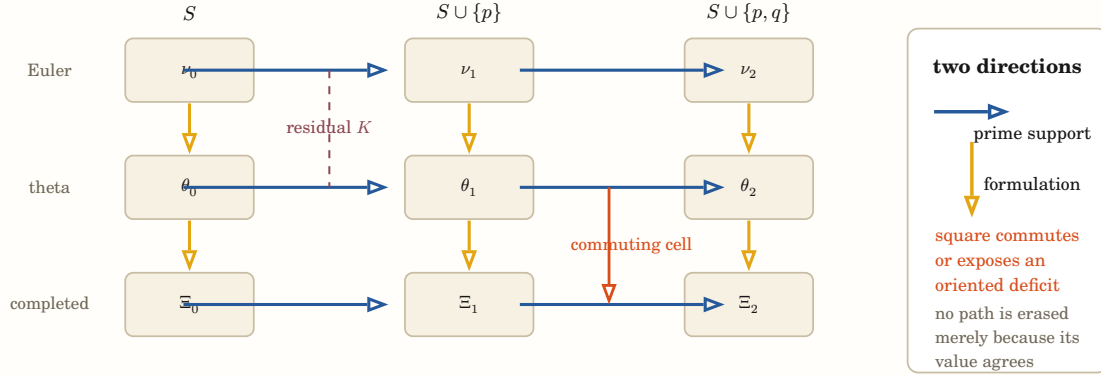


Figure 7: The proposed formulation atlas is fibered in two directions. Horizontal maps enlarge or transfer finite prime support; vertical maps change the formulation receiver. A commuting square certifies one exact translation. A noncommuting square records the explicit residual needed to extend the atlas; it is not silently identified with zero.

A structural two-cell is a map between the apexes of two parallel spans which commutes with both legs and preserves the declared decorations. When the receiver is additive, the quantity

$$K = U_2 \circ U_1 - T_2 \circ T_1$$

may be recorded as a receiver residual. $K = 0$ testifies that the selected receiver square commutes; it does not by itself reconstruct a map of complete interiors. $K \neq 0$ is an informative OPEN, not a fabricated two-cell: it identifies precisely which relation a larger atlas would have to supply.

The whole atlas is therefore not a groupoid. Only reversible rechartings and invertible comparison cells form a bigroupoid; after invertible 2-isomorphic arrows are identified, that core gives the ordinary rechart groupoid. Parameter pullback may organize this core into a category fibred in groupoids, but only after its cartesian lifts and coherence are actually constructed.

VI.2 Compression in the atlas

Compression is not the deletion of the path because its endpoint is known. It is the outgoing face which factors inactive interior directions while retaining the consequential relation needed by later transport.

For example, the scalar $\frac{\pi}{4}$ is a valid returned face of the Machin cell. The exact identity remains able to re-open into Gaussian factors, arguments, norms, and prime axes 2, 13. A decimal approximation which retains none of those relations is a narrower receiver quotient. It can be useful, but it is not the same informational face.

VII Ray tracing as a situated mathematical receiver

VII.1 Why the gallery is relevant

MathWorld places the requested objects under “Mathematical Art” [9]. Their visual interest is not itself a theorem, and human salience is not an invariant. Yet this particular gallery is heavily enriched for compact equations with unusually structured global consequences:

- extremal or repeated singularities;
- large symmetry groups and quotient tilings;
- tangent-packing or reflection orbits;
- minimal or variational surfaces with period closure;
- algebraic level sets which change topology at discriminants; and
- intersections whose visible branches depend strongly on the chosen chart.

The image is informative because the underlying relations are constrained, not because pleasant rendering confers mathematical truth.

VII.2 The camera cannot leave the system

Definition 9 — Situated ray receiver of an implicit geometry Let an intrinsic real affine face be given by

$$X_F = \{x \in \mathbb{R}^n : F(x) = 0\}.$$

A ray receiver is the declared tuple

$$R = (o, d, I, \chi, L),$$

where o is its source, d its oriented direction, I the admissible parameter interval, χ its root-selection and occlusion law, and L its lighting or measurement law. It observes roots of

$$F(o + td) = 0, \quad t \in I,$$

selects a visible occurrence t_R through χ , and may evaluate the regular normal $\nabla F(o + t_R d)$. The rendered face is therefore

$$\Pi_{R(X_F)} = L(o, d, t_R, \nabla F(o + t_R d), \dots),$$

whenever these terms exist.

Boundary. $\Pi_{R(X_F)}$ is not the whole intrinsic geometry. It depends on an affine chart, a real slice, source, direction, root selection, occlusion, and measurement law. At a singular occurrence $F = 0$ and $\nabla F = 0$ the regular normal is unavailable; the singularity belongs to the intrinsic discriminant rather than being a shading defect.

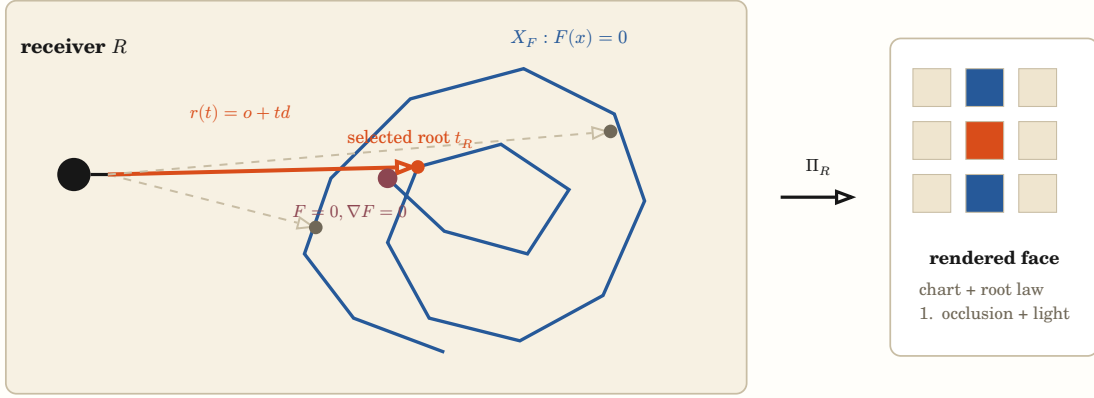


Figure 8: Ray tracing is itself a receiver. It intersects one declared real affine chart along $r(t) = o + td$, chooses an admissible visible root, and maps local differential data into pixels. The result can faithfully testify about that cut while still hiding complex points, projective closure, interior sheets, occluded roots, and singular resolutions.

An implicit surface $F(x) = 0$ can intersect one ray at several values of t . The visible image chooses an interval, a root order, and an occlusion law. It then maps local incidence and differential data to color. When

$$F(x) = 0, \quad \nabla F(x) = 0,$$

the ordinary normal is not defined: the intrinsic surface has reached its singular stratum. Shading can disclose the neighborhood, conceal it, or render an artifact, but it does not create the singularity.

This is the holomorphic correction to the “absolute camera.” A useful renderer may imitate an outside Euclidean viewpoint, but that viewpoint is implemented by a lawful relation between source, rays, affine chart, surface, root selection, light, and pixel receiver.

VIII Four surface cells

VIII.1 Bowl of Integers: an integer orbit of tangent constraints

The Bowl of Integers is an Apollonian sphere packing whose tangent spheres carry integral bends [10]. In three dimensions, five mutually tangent oriented spheres with bends $b_1, \dots, b_5$ satisfy the Soddy–Gossett relation [11]:

$$\left(\sum_{j=1}^5 b_j \right)^2 = 3 \sum_{j=1}^5 b_j^2.$$

Hold four bends $a_1, \dots, a_4$ fixed. The two possible fifth bends x, x' are roots of the same quadratic, hence

$$x + x' = \sum_{j=1}^4 a_j.$$

Replacing x by

$$x' = \sum_j a_j - x$$

is an integer-preserving reflection. Repeated reflections grow the packing. The observed integer sequence is therefore an orbit of a local tangency law, not a list laid into space after the fact.

Factoring the bends gives prime-valuation faces of this orbit. Those factors can classify recurrence in selected receivers, but they do not replace the tangency relation which causes the next sphere. This is the same distinction the prime atlas makes between an address and a transport law.

VIII.2 Cayley cubic: singular nodes as founding seams

The Cayley cubic is the unique cubic surface with the maximal four ordinary double points; it has tetrahedral symmetry and nine lines [12], [13]. One projective equation is obtained by clearing denominators in

$$\frac{1}{x_0} + \frac{1}{x_1} + \frac{1}{x_2} + \frac{1}{x_3} = 0.$$

The four nodes are locations where the regular local surface law fails:

$$F = 0, \quad \nabla F = 0.$$

They are discriminant seams, not empty points. A resolution replaces each collapsed node by exceptional geometry which records limiting directions.

A ray-traced real affine image shows one receiver face of the cubic. It does not show every projective line, every complex point, or the exceptional curves of a resolution. The compelling “kink” is thus a visible witness of an intrinsic rank failure, while the full identity of that seam belongs to the larger atlas.

VIII.3 Chen–Gackstatter: local complex data and global period closure

The Chen–Gackstatter surfaces form a double-indexed family M_{ij} of complete minimal surfaces [14], [15]. In the convention reported by MathWorld, the genus is ij , the total curvature is

$$-4\pi(i+1)j,$$

and the single Enneper-type end winds $2j+1$ times.

Their Weierstrass data are local complex information. Integration produces a real minimal immersion only when the relevant periods close. This gives the atlas a rigorous model of the laboratory’s “string winding” intuition: local differential data can be valid at every step while a global loop still returns a nonzero period. The period residual is exactly what must vanish or be quotiented for the surface to close.

The indices do not merely scale one picture. They change genus, winding, and total curvature discretely. The family is a parameterized topology in which continuous local transport and quantized global phase coexist.

VIII.4 Klein quartic: the strongest prime–archimedean surface cell

The Klein quartic has complex projective equation

$$x^3y + y^3z + z^3x = 0$$

and genus 3. It is also a compact hyperbolic quotient associated with level 7 and has orientation-preserving automorphism group

$$\text{PSL}(2, 7)$$

of order 168 [16], [17]. Its regular map has 24 heptagonal faces, 84 edges, and 56 vertices:

$$56 - 84 + 24 = -4 = 2 - 2 \cdot 3.$$

The counts share the finite-prime support

$$24 = 2^3 \cdot 3, \quad 84 = 2^2 \cdot 3 \cdot 7, \quad 56 = 2^3 \cdot 7, \quad 168 = 2^3 \cdot 3 \cdot 7.$$

The prime 7 selects the congruence level and heptagonal local incidence; the upper half-plane supplies archimedean hyperbolic geometry; the finite group supplies symmetry transport; the algebraic equation supplies a complex-projective face; and the rendered tiling supplies a situated visual receiver. These are genuinely different formulations of one object, joined by nontrivial mathematics. This is the clearest surface prototype for the prime–archimedean atlas.

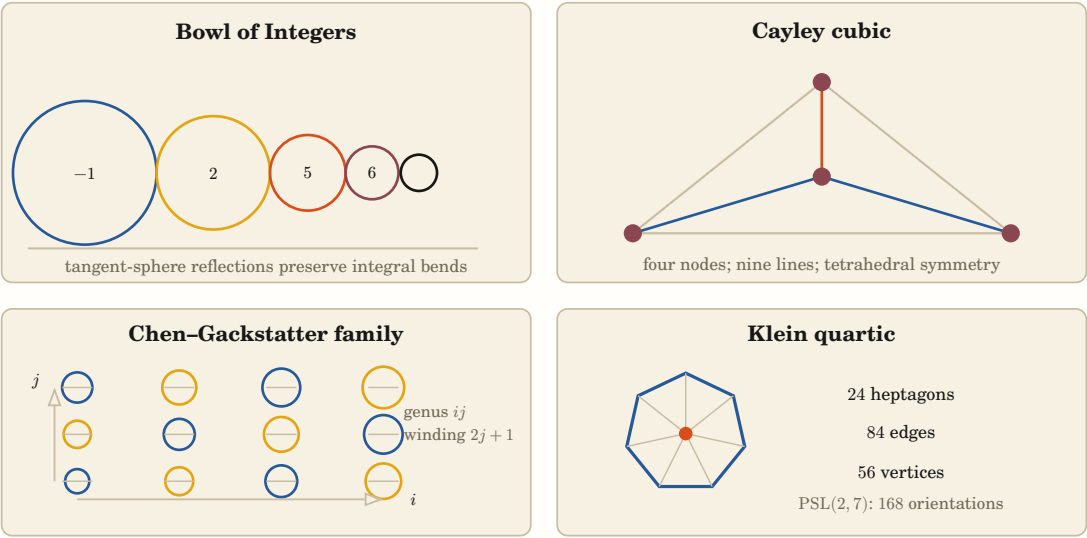


Figure 9: Four original incidence schematics, deliberately not photorealistic. Each ray-traced subject is compelling because a compact local law unfolds into a globally constrained population: reflection orbits, extremal singularities, period-closed minimal surfaces, and a hyperbolic quotient of exceptional symmetry.

IX What the rest of the gallery contributes

The four requested objects do not exhaust the useful families. The gallery can be reorganized by the law which produces global complexity:

Atlas family	Examples	What becomes visible
Packing / contact	Bowl of Integers, bubbles, kissing configurations, spherical codes, Reuleaux tetrahedron	local tangency or distance constraints propagated into a global contact graph and reflection orbit
Algebraic discriminant	Cayley, Chmutov, Clebsch, Endrass, Hunt, Kummer, Sarti, Cassini	singular points, exceptional line populations, nodal extrema, and phase changes of real sections
Variational / minimal	Chen–Gackstatter, bubble and double-bubble surfaces	local stationarity constrained by global period, boundary, and topology
Quotient / cover	Klein quartic, highly symmetric cubic and quartic surfaces	many local tiles identified under a finite or Fuchsian group action
Intersection / envelope	cylinder–sphere, miter, implicit chair, cushion, heart, piriform	receiver-dependent branch selection, self-intersection, and changing cross-section

Three further examples sharpen the point:

- A generic Kummer quartic has the maximal 16 ordinary double points and a 16_6 configuration of nodes and tropes [18]. The visible nodal population is the shadow of a tightly constrained incidence system.
- Chmutov surfaces use Chebyshev recurrences to produce many ordinary double points [19]. A one-dimensional recurrence is lifted into a high-complexity algebraic surface.
- Cassini surfaces and their planar sections are product-distance level sets whose real topology changes as the level crosses a discriminant [20]. One formula supports several phases separated by an exact boundary.

The shared lesson is not that every attractive shape is a prime theorem. It is that local laws, repeated under constraints and then quotiented through a receiver, can generate complex faces whose seams disclose the topology that the receiver cannot flatten away.

This is also the precise connection to fractals. A Julia set is generated by iterating one rational map in its dynamical plane; the Mandelbrot set classifies parameter values by the behavior of the corresponding critical orbit. The parameter plane and dynamical plane are nonidentical atlas faces, and an escape-time image is a receiver of the orbit rather than the orbit itself [21], [22]. The gallery surfaces need not be fractals to share the important mechanism: a compact law is recurrently applied, critical or singular seams organize phase, and a projection exposes only one cross-section of the resulting topology.

X Relation to the Riemann Hypothesis

X.1 What this atlas adds to the existing RH paper

The laboratory’s receiver-geometry paper already isolates the classical spine:

Euler product $\rightarrow$ theta / Poisson $\rightarrow$ Mellin / Gamma completion $\rightarrow$ explicit formula / Weil response.

The present atlas adds five things.

First, the finite-prime side now has an exact combinatorial carrier: valuation simplices, root-lattice moves, residue fibers, prime-power apertures, and the weighted current $\psi(X)$.

Second, the vertical direction is no longer “some other formula.” It has populated cells: Gaussian argument/norm transport for Machin’s formula, character transport for the π Euler product, factorial valuation transport for e , exponential covering transport for $e^{i\pi}$, and Gaussian–theta–Mellin transport at the infinite place.

Third, observation is typed. A planar plot, ray trace, or scalar evaluation is a receiver quotient with a declared root, branch, chart, and boundary. It may reveal a singularity or symmetry orbit, but it cannot stand in for the unobserved complex or projective geometry.

Fourth, the fixed seam now has exact cross-chart conduct. The same normal coordinate $\varepsilon = \text{Re}(s) - \frac{1}{2}$ is retained by the completed involution, logarithmic rebase, reciprocal spiral pair, centered Smith residual, and rational sphere lift. A regular transformed-grid crossing, a zero, a critical branch, and a distinct-source coincidence remain different events throughout that transport.

Fifth, the reciprocal sphere pair now has a discrete connection carrier. The complex-plane intersection, axis orientation, two pole paths, common metric, coupling metric, conic discriminant, and triangular holonomy remain separately readable. Consequently an observed kink can be classified rather than treated as generic visual complexity.

X.2 Why infinitely many π – e formulas matter

The formulas do not prove RH by numerical abundance. They matter because they furnish a large family of exact transition cells in which:

- the returned invariant is known;
- the causal formulation differs;
- finite prime axes can be calculated;
- archimedean functions such as $\exp$, $\sin$, Γ , elliptic integrals, and Gaussian kernels intervene;
- branch, convergence, and period boundaries are explicit; and
- failures to commute can be localized rather than averaged away.

This is a practical training ground for the much harder completed-zeta atlas. It asks whether recurrent transformation species can be classified without pretending there is one absolute formula for the constant.

X.3 The local equivalence and the global carrier

Let

$$\rho = \frac{1}{2} + \varepsilon + i\gamma$$

be a symbolic nontrivial completed-zero occurrence. Conditional on $\Xi(\rho) = 0$, the new atlas carries the exact local equivalence

$$\varepsilon = 0 \quad \text{iff} \quad \rho \in \text{Fix}(J) \quad \text{iff} \quad \rho_{\varepsilon, \gamma} \text{ is unitary} \quad \text{iff} \quad |W_{a(\varepsilon+i\gamma)}| = 1.$$

The usual sentence “the real part is $\frac{1}{2}$ ” is therefore one projection of a richer geometric claim: the completed zero has no normal growth or decay under logarithmic rebase and no normal displacement from the centered Smith shell. If $\varepsilon \neq 0$, completion instead supplies reciprocal zero partners on two opposed spiral/sphere hands.

The moving-chord cell gives those hands worldlines. It shows exactly how one plane cut can become asymmetric while the chord remains rigid, how translation couples to precession, and how a triangular loop can retain holonomy. It does not yet say which such paths are Ξ paths. That distinction is now the global boundary rather than an ambiguity in the carrier.

The unresolved edge is global rather than visual. One law must rule out those off-shell nontrivial zero modes for every admitted completed current, not only for selected heights or after choosing a favorable orientation. The existing receiver-geometry work expresses that law through the completed Hermitian response.

Let $\mathcal{H}_\lambda$ be the admitted test-current fiber at an atlas configuration λ , with completed Hermitian response

$$W_{\lambda(f,g)}.$$

For a transition path γ with transport T_γ , define the response residual

$$\mathcal{K}_{\gamma(f,g)} = W_{\gamma(1)}(T_\gamma f, T_\gamma g) - W_{\gamma(0)}(f, g).$$

An RH-bearing atlas cannot merely find, for each current, some orientation in which a plotted scalar looks positive. It must establish an anchored cover in which:

1. every admissible completed current is represented;
2. transitions carry the same completed occurrence;
3. adjoints and boundary conditions transport covariantly;
4. the Hermitian signature is preserved, or every residual is independently controlled; and
5. holonomy around every admitted loop cannot turn a positive direction into an unaccounted negative one.

In one fixed Weil chart, universal nonnegativity of the completed response is equivalent to excluding the off-shell nontrivial modes. The formulation atlas must therefore establish coverage and covariance: it must carry the same response through a rechart, rather than creating positivity by reorienting each occurrence after observation. In the moving-chord chart, this asks for a pullback identifying the completed response with the two-sheet metric and connection. Only such an identity can turn “normal chord vanishes” from a local equivalence into a consequence for every genuine zero mode.

X.4 The next global square

The smallest meaningful next square is:

$$\text{Path A: } C_S \xrightarrow{\text{add } p} C_{S \cup \{p\}} \xrightarrow{\text{complete}} \hat{C}_{S \cup \{p\}},$$

$$\text{Path B: } C_S \xrightarrow{\text{complete}} \hat{C}_S \xrightarrow{\text{transport } p} \hat{C}_{S \cup \{p\}}.$$

The earlier RH paper derived the local Euler metric deformation and its prime-power recurrence. What remains is to calculate the vertical theta–Mellin transport of that complete successor cell, including its archimedean and aperture-connection terms, and then determine the signature of the residual. The π – e atlas gives exact exemplars for how such a cell must retain prime factors, branch, scale, and returned invariant simultaneously.

XI Bounded analytical program

The bounded presentation-topology construction has populated the horizontal term and accumulation paths and one vertical Machin–Gaussian support change. The conformal-rebase construction has now populated the local parameter, germ, completion, rebase, and Smith fibers. The next work is to carry one complete occurrence between nonidentical formulations, not enumerate more primes or zero coordinates until a pattern appears.

Progress	Construction	Required observable
1 · populated	Series presentation topology	Exact factor channels, recurrence annihilation, additive compression, moving carriers, semiprime faces, and complete event paths for three arctangent arms and Chudnovsky.
2 · populated locally	Machin vertical cell	Exact passage from the {5, 239} series-scale chart to the {2, 13} Gaussian norm chart, including orientation and alternating-tail bounds.
3 · populated locally	Conformal parameter/rebase atlas	Exact germ degree, zero and critical orders, source-preserving coincidences, completed fixed/normal projectors, reciprocal log-polar rebase, and centered Smith-shell residual.
4 · populated locally	Precessing two-sheet carrier	Exact moving Smith chord, plane intersection, pole-edge sum and difference laws, common and coupling metrics, flat-versus-curved triangular connection, and one conic discriminant path.

Progress	Construction	Required observable
5 · next	Cross-presentation transport	Carry one complete divisor-and-boundary path between nonidentical formulations. Compare the two composites and localize any residual to an explicit seam or holonomy.
6 · open	Zeta completion response cell	Carry one finite Euler successor through Gaussian, Poisson, and Mellin transformations; derive its two-sheet metric and connection from the completed field; then evaluate its Hermitian response, aperture terms, and loop holonomy.

The qualitative success signature is not “a beautiful image.” It is a nontrivial square whose two complete paths agree, or whose disagreement is localized to an interpretable boundary term. The stopping condition is the first such completed cell or the first exact obstruction.

The three exact constructions are deposited at `observations/eros-transcendental-presentation-topology-01/RESULTS.md` and `observations/eros-conformal-rebase-atlas-01/RESULTS.md`, and `observations/eros-precressing-chord-conic-transport-01/RESULTS.md`. The remaining synthesis regrades, rather than reruns, the exact records at `observations/prime-spectral-world-01/RESULTS.md`, `observations/prime-fraction-navigation-01/RESULTS.md`, `observations/eros-formula-ecology-01/RESULTS.md`, `observations/eros-formula-aperture-growth-01/RESULTS.md`, and `observations/eros-theta-mellin-aperture-01/RESULTS.md`. Their finite apertures remain finite; the atlas supplies the common mathematics in which their complete path populations can now be compared.

XII Conclusion

The prime–archimedean formulation atlas now has an exact mathematical skeleton.

Finite prime composition lives in the lattice $\mathbb{Z}^S$ and its nonnegative graded simplices. Prime-axis transfers form A_{r-1} ; fractions carry oriented valuation displacement; analytic specialization sends those moves to exact complex amplitude ratios. Residue loops and primordial wheels are separate fibers connected by reduction, not substitutes for valuation.

The constant formulations populate the vertical direction. Machin’s formula closes through a Gaussian integer identity whose norms expose axes 2 and 13. A character Euler product relates π to residue orientation of every odd prime. The factorial series for e traverses prime powers by Legendre’s law, while continued fractions, differential flow, and the exponential cover provide nonidentical causal faces. Euler’s identity is a covering-space seam, not the collapse of line and loop into an untyped symbol.

The bounded construction now shows what this means during evaluation rather than only at the formula endpoints. An arctangent term carries a moving odd factor against a persistent scale rail; its partial sum carries a second, coupled accumulation topology. Chudnovsky has the same causal form at higher arity, with factorial lifts, a fixed prime-support scale, and a linear factor that enters, returns, and departs. Exact recurrence annihilation and exact additive compression are different events. Machin’s Gaussian closure then proves that a lawful formulation change can replace the local finite-prime support while preserving the returned angle.

Situated mean transport supplies the continuous half of this geometry. Along an admitted Riemannian path, a scalar change is the integral of its directional differential and has a path-relative mean witness. Sections in different fibers require parallel transport; alternative routes can return holonomy. Branch,

rank, phase, and valuation seams remain explicit discrete terms. This is the rigorous relation between continuous deformation and the quantized events already visible in the exact series paths.

The exact product formula already joins every finite place of a rational to its archimedean magnitude. Zeta completion lifts this relation into the Gaussian–theta–Poisson–Mellin chain. The critical line is the fixed locus of the completed functional-equation reflection.

The conformal-rebase cut now carries that fixed locus through three further faces. Local germ degree distinguishes a regular crossing, zero, critical branch, and distinct-source coincidence before rendering. The centered normal ε becomes the real exponent of logarithmic rebase and the signed Smith-shell displacement. Its two nonzero hands are reciprocal spirals and opposed sphere normals; at $\varepsilon = 0$ they close to a constant-radius orbit and the unit shell. Thus “real part $\frac{1}{2}$ ” is an exact fixed/normal factorization with preserved cross-chart conduct, not an absolute line drawn over a plane.

The moving-chord cut then gives the reciprocal hands an exact discrete geometry. A rational tilt changes the complex-plane intersection without changing the rigid chord; rational precession transports both that intersection and the axis direction by one complex phase. The two pole edges separate common activity from oriented coupling, their induced sheet metrics retain the same distinction, and triangular holonomy distinguishes flat closure from a curved return. A conic discriminant remains a separate phase seam rather than being collapsed into either curvature or intersection.

The ray-traced gallery then becomes mathematically relevant without becoming ornamental evidence. The Bowl of Integers is an integer reflection orbit of tangency constraints. The Cayley cubic concentrates topology at extremal singular seams. Chen–Gackstatter surfaces turn local complex data into a global period problem with discrete genus and winding. The Klein quartic joins the prime level 7, the finite group $\text{PSL}(2, 7)$, hyperbolic archimedean geometry, algebraic form, topology, and a rendered quotient in one especially complete atlas cell. Their images are situated projections of those relations.

The RH-bearing object is now sharply bounded. Local parameter transport already distinguishes regular tangency, zero order, branch degree, normal hand, reciprocal rebase, shell incidence, moving pole paths, induced sheet metric, and connection holonomy. One complete divisor-and-boundary path must next be carried between nonidentical formulations. That supplies the lawful vertical transport needed to carry a finite-prime successor through archimedean completion and calculate the Hermitian response and holonomy of the resulting square. The decisive identity must derive that response on the two-sheet carrier and force its normal zero-mode component to vanish. The square may close or expose an exact boundary residual without an absolute picture or an exhaustive prime search.

Bibliography

- [1] D. H. Bailey, “A Catalogue of Mathematical Formulas Involving Pi, with Analysis,” 2021, [Online]. Available: <https://www.davidhbailey.com/dhbpapers/pi-formulas.pdf>
- [2] E. W. Weisstein, “Pi Formulas.” Accessed: Jul. 23, 2026. [Online]. Available: <https://mathworld.wolfram.com/PiFormulas.html>
- [3] E. W. Weisstein, “e.” Accessed: Jul. 23, 2026. [Online]. Available: <https://mathworld.wolfram.com/e.html>
- [4] E. W. Weisstein, “Cauchy’s Mean-Value Theorem.” Accessed: Jul. 23, 2026. [Online]. Available: <https://mathworld.wolfram.com/CauchysMean-ValueTheorem.html>
- [5] B. Conrad, “Covariant Derivative, Parallel Transport, and General Relativity.” Accessed: Jul. 23, 2026. [Online]. Available: <https://math.stanford.edu/~conrad/diffgeomPage/handouts/covariantder.pdf>
- [6] “Parallel Transport.” Accessed: Jul. 23, 2026. [Online]. Available: https://encyclopediaofmath.org/wiki/Parallel_transport
- [7] NIST Digital Library of Mathematical Functions, “Chapter 20: Theta Functions.” Accessed: Jul. 23, 2026. [Online]. Available: <https://dlmf.nist.gov/20>
- [8] NIST Digital Library of Mathematical Functions, “Chapter 25: Zeta and Related Functions.” Accessed: Jul. 23, 2026. [Online]. Available: <https://dlmf.nist.gov/25>
- [9] E. W. Weisstein, “Ray-Traced Images.” Accessed: Jul. 23, 2026. [Online]. Available: <https://mathworld.wolfram.com/topics/Ray-TracedImages.html>
- [10] E. W. Weisstein, “Bowl of Integers.” Accessed: Jul. 23, 2026. [Online]. Available: <https://mathworld.wolfram.com/BowlofIntegers.html>
- [11] T. Gossett, “The Kiss Precise,” *Nature*, vol. 139, p. 62, 1937, doi: [10.1038/139062a0](https://doi.org/10.1038/139062a0).
- [12] E. W. Weisstein, “Cayley Cubic.” Accessed: Jul. 23, 2026. [Online]. Available: <https://mathworld.wolfram.com/CayleyCubic.html>
- [13] E. W. Weisstein, “Ordinary Double Point.” Accessed: Jul. 23, 2026. [Online]. Available: <https://mathworld.wolfram.com/OrdinaryDoublePoint.html>
- [14] C. C. Chen and F. Gackstatter, “Elliptische und hyperelliptische Funktionen und vollständige Minimalflächen vom Enneperschen Typ,” *Mathematische Annalen*, vol. 259, pp. 359–369, 1982, doi: [10.1007/BF01456947](https://doi.org/10.1007/BF01456947).
- [15] E. W. Weisstein, “Chen-Gackstatter Surfaces.” Accessed: Jul. 23, 2026. [Online]. Available: <https://mathworld.wolfram.com/Chen-GackstatterSurfaces.html>
- [16] S. Levy, *The Eightfold Way: The Beauty of Klein’s Quartic Curve*. Cambridge University Press, 1999.
- [17] E. W. Weisstein, “Klein Quartic.” Accessed: Jul. 23, 2026. [Online]. Available: <https://mathworld.wolfram.com/KleinQuartic.html>
- [18] E. W. Weisstein, “Kummer Surface.” Accessed: Jul. 23, 2026. [Online]. Available: <https://mathworld.wolfram.com/KummerSurface.html>
- [19] E. W. Weisstein, “Chmutov Surface.” Accessed: Jul. 23, 2026. [Online]. Available: <https://mathworld.wolfram.com/ChmutovSurface.html>
- [20] E. W. Weisstein, “Cassini Surface.” Accessed: Jul. 23, 2026. [Online]. Available: <https://mathworld.wolfram.com/CassiniSurface.html>
- [21] E. W. Weisstein, “Julia Set.” Accessed: Jul. 23, 2026. [Online]. Available: <https://mathworld.wolfram.com/JuliaSet.html>
- [22] E. W. Weisstein, “Mandelbrot Set.” Accessed: Jul. 23, 2026. [Online]. Available: <https://mathworld.wolfram.com/MandelbrotSet.html>